# Pippi de Bree

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## Education

**New York University, Steinhardt** **Expected May 2027**
Master of Science in Applied Statistics for Social Science Research
GPA: 4.0

**BrainStation Data Science Bootcamp, London, UK** **November 2022**
Admitted with a full scholarship to twelve week bootcamp.

**Boston University, College of Arts and Sciences** **January 2020**
Bachelor of Arts in Economics and Mathematics, Minor in Computer Science
Dean's List Spring 2018, Fall 2018 and Spring 2019

## Peer Reviewed Publications

**de Bree, P.**, Mesalles, V., Morelli, A., Downey, D., Wang, X., Chouthai, M., Bachand-Bergeron, M., Park, J., Thakral, A., Lepore, N., Harel, D. Don't worry, I asked ChatGPT: Investigating use of Generative artificial intelligence among sexual and gender minorities and cisgender heterosexual individuals. (In preparation for review)

Scott, M., **de Bree, P.**, Vandewalle, V., Gauthier, JA., Le Goff, JM. What Counts? The Use of History in Sequence Analysis and Prediction (Under Review) *Sociological Methodology*

Scott, M., Jia, B., **de Bree, P.**, Gauthier, JA., Le Goff, JM. The Branching Tree of the Life Course: Incorporating History in Sequences Analysis (In preparation for review)

## Research Experience

**Research Assistant** **Summer 2026 - Present**
**Supervisor: Dr. Klint Kanopka**

Sparse Variational Autoencoders for Item Response-Like Modeling

- Self-led investigation into adapting a deep generative model to Multidimensional Item Response Theory. Adapted model to recover interpretable parameters.
- Built three exploratory variational autoencoders in PyTorch.
- Evaluated efficacy of sparse decoding (through spike and slab priors) to identify models in space, using paranormal beliefs data.
- Tuned priors and created training visualizations for parameter interpretation.
- Authored a comparison about the relationship between this novel method and traditional psychometric models.

**NYU QUEER Data Lab, Research Assistant** **Fall 2025 - Present**
**Supervisor: Dr. Daphna Harel**

Don't worry, I asked ChatGPT: Investigating use of Generative artificial intelligence among sexual and gender minorities and cisgender heterosexual individuals.

- Led statistical analyses and evaluation of methods to derive perceptions from an AI use and sentiment survey, wrote the code to do so in R.
- Conducted a statistical literature review of modeling options and communicated the implications to the lab, leading the selection of the final modeling strategy.
- Fit factor analyses and a multi-level model to compare estimated marginal mean uses and perceptions of AI across sexual and gender minority+ groups.
- Collaboratively wrote the statistical methods, results, and discussion sections.

**Research Assistant** **Fall 2025 - Present**
**Supervisor: Dr. Marc Scott**

What Counts? The Use of History in Sequence Analysis and Prediction

- Set up cross validation paradigms to evaluate a novel feature creation method, using categorical functional data analysis, against traditional methods for sequence analysis.
- Methods included hard and soft clustering, using sequence-based distance metrics, principal component analysis and established sequence metrics features.
- Built the framework for a simulation study as well as benchmarked real-world data.
- Performed sensitivity analyses on all methods to ensure a "fair" competition. Tuned parameters and built a test set assignment function for soft clustering.
- Developed visualizations to compare time-varying representations of sequences.
- Wrote the Methods section and several appendices regarding sensitivity analyses, collaborated with Professor Scott on writing the remainder of the paper.

The Branching Tree of the Life Course: Incorporating History in Sequences Analysis

- Fit hard and soft clustering, with sequence-focused distance metrics, on subsequences of school-to-work transition data for comparison against a method of modeling multi-phase latent classes, utilizing model-based clustering.
- Developed visualizations of state probabilities, across time for comparison against visualizations from model based clustering with transition points method.
- Wrote the appendix describing the distance based clustering method.

## Course Projects

**Can LLMs Compare to Human Content Moderators?** **Fall 2025**
Messy Data and Machine Learning | NYU Course Project for APSTA-GE 2047

- Performed the technical and statistical work of a group project comparing LLM against crowd-sourced ratings of content moderation on X's Community Notes.
- Created a subset of conclusive content moderation notes, and corresponding ratings.
- Developed a prompting pipeline, in R, to generate synthetic content moderation ratings from the LLM. Iterated over prompts to ensure a strict form of output data, to allow for modeling.

- Fit random forest and logistic regression models on human-ratings training data to benchmark the relationship between ratings survey variables and outcomes. Compared performance of test set of LLM survey responses against corresponding withheld human-ratings set.
- Explored limitations of an overfit model due to large differences found in LLM survey responses.

**Exploring Recommendation System Suitability by User History Depth** **Fall 2022**
Capstone | BrainStation

- Built four recommendation systems in Python to compare users with varying levels of listening history. Recommenders included a popularity based, content based and two collaborative filtering systems based on latent feature processing.
- Presented findings in an interactive presentation with real-time recommendations.

## Professional Experience

**Quantitative Student Specialist | NYU Data Services** **May 2026 - Present**

- Provide cross-disciplinary quantitative research consultations to members of the NYU community, focusing on statistical software support (R, Stata) as well as data cleaning and manipulation.
- Lead R coding tutorial "Data Wrangling in R". Adapted content to modern R practices.
- Collaborate with subject area specialists and data librarians to support research data lifecycle.

**Machine Learning Engineer Intern | Durable** **Spring-Summer 2023**
Louisville, CO

- Built a Retrieval Augmented Generation model by customizing code of pre-trained Hugging Face encoders and generators.
- Developed pre-training for the retrieval system to ensure encodings took on bespoke language form necessary for integration with code synthesis paradigm. Created bespoke evaluation metrics for pre-training based on the retriever's ranking.
- Designed a training pipeline, using PyTorch Lightning and ClearML reporting.
- Collaborated closely with the full start-up team, of 6, to integrate the model into the product.

## Teaching

**Teaching Assistant | NYU APSTA-GE 2047** **Fall 2026**
Messy Data and Machine Learning

- Scaffolded curriculum for recitation sessions throughout the semester. Wrote lab sessions on training neural networks using torch in R.
- Led recitation sections for 34 graduate students and held office hours to discuss concepts and homework problems.
- Consulted on group project planning and modeling feasibility.

- Set up batch homework autograding for coding problems and hand graded written responses.

**Teaching Assistant | NYU APSTA-GE 0021** **Spring 2026**
Cracking the Code

- Developed and led an Introduction to R programming lecture for non-technical undergraduate students.
- Led a lecture and lab sessions teaching students to use the programming language R, in RStudio.
- Held office hours to discuss coding projects, concepts and homeworks.

## Coursework

**Graduate**
Frequentist Inference
Bayesian Inference
Causal Inference
Multi-Level Modeling
Supervised and Unsupervised Machine Learning
Messy Data and Machine Learning
Generalized Linear Models
Data Science for Social Impact

**Undergraduate**
Linear Models
Probability
Linear Algebra
Differential Equations
Combinatronic Structures
Behavioral Economics
Game Theory
Empirical Econometric Analysis I&II

## Skills & Technologies

Advanced R (base, tidyverse, ggplot)
Advanced Python (Pandas, Numpy, Scikit-learn)
PyTorch, PyTorch Lightning, SQL, Git, Stata